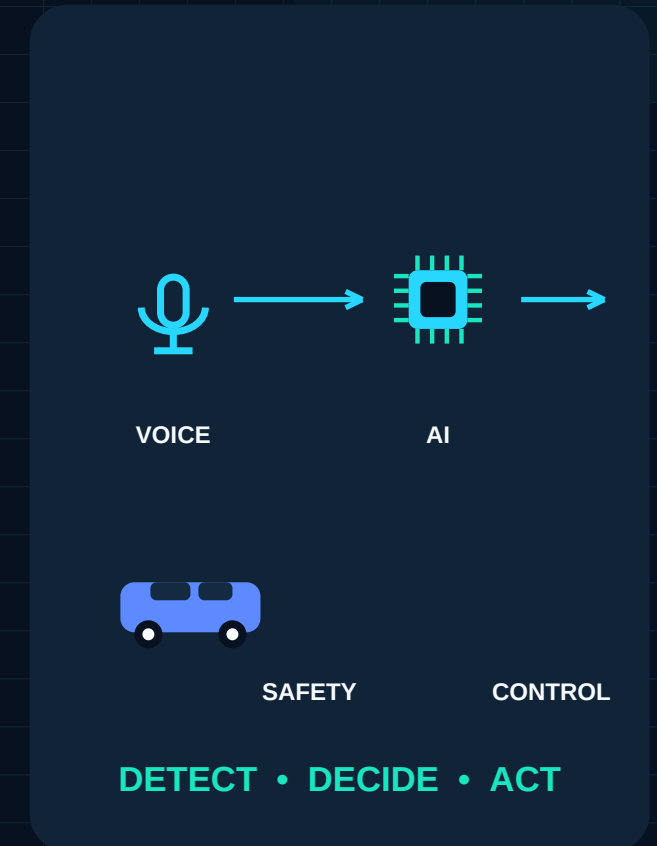


Emotion-Aware Adaptive Vehicle Control Using On-Device AI

From driver emotion → intelligent safety decision → real-time vehicle response



NO CAMERA

ON-DEVICE AI

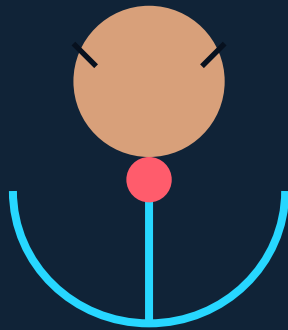
REAL-TIME

Prototype platform: Snapdragon-powered HP PC + wireless embedded vehicle controller

01 | THE PROBLEM

A safety gap between driver state and vehicle response

Driver state can change
before the vehicle responds.



VOICE + HUMAN STATE

01

Stress / anger can affect driving behaviour

02

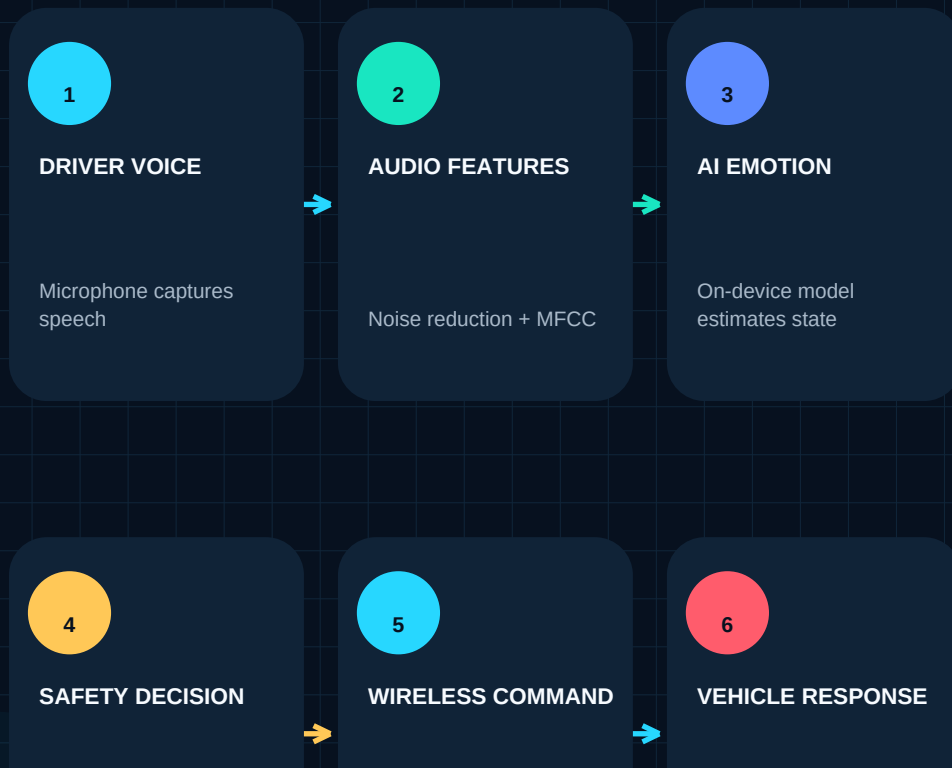
Traditional monitoring often relies on cameras

THE GAP

Detecting emotion is useful only when the system can respond safely.

02 | OUR SOLUTION

A closed-loop system that turns emotion into action



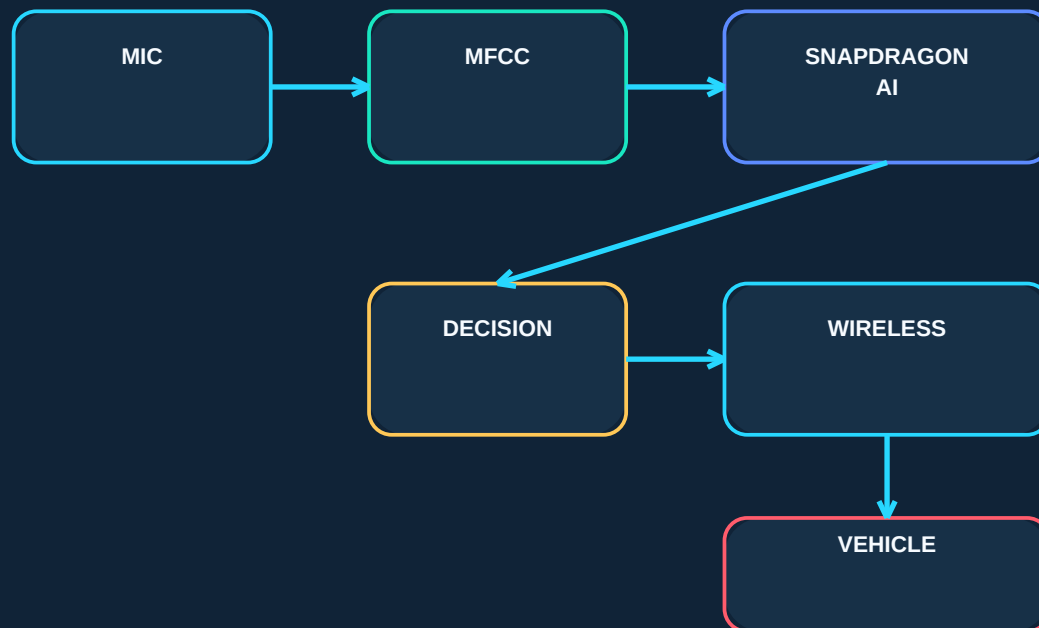
EMOTION → DECISION → VEHICLE ACTION

The project goes beyond emotion recognition by connecting AI output to a physical safety response.

03 | SNAPDRAGON AI FIT

The HP Snapdragon PC acts as the AI brain in the prototype

Prototype Architecture

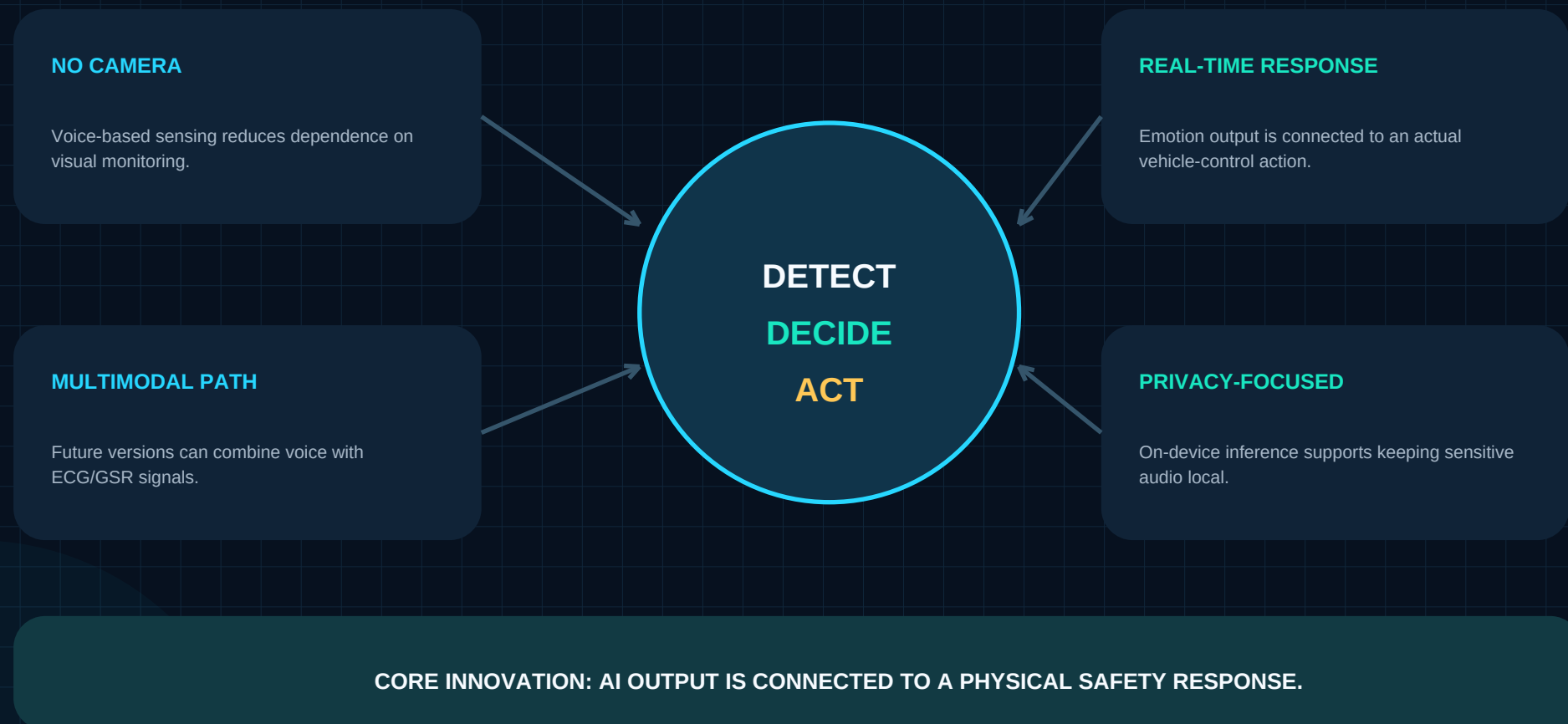


Why On-Device AI?

- **LOW LATENCY**
Local inference supports a responsive prototype.
- **PRIVACY**
Voice data need not be continuously sent to a cloud service.
- **EDGE READY**
AI processing happens close to the application.
- **CHALLENGE FIT**
Architecture is designed around a Snapdragon-powered HP PC.

04 | WHAT MAKES IT DIFFERENT

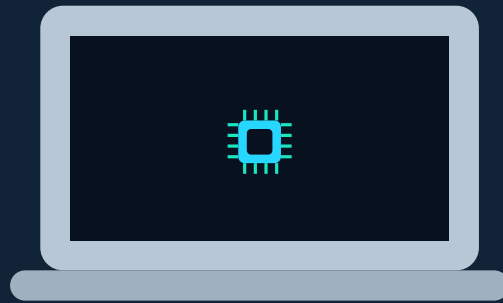
One idea, connected end-to-end



05 | DEMO, IMPACT & NEXT STEP

A simple prototype today — a path toward embedded automotive systems

Live Demo Concept



SNAPDRAGON-POWERED HP PC



ARDUINO VEHICLE

Speak → detect emotion → send command →
observe vehicle speed response.

Expected Demonstration Value

FAST

Local AI inference

SMART

Emotion-aware decision

PHYSICAL

Real vehicle response

SCALABLE

Path to embedded automotive hardware

NEXT: optimize the emotion model for Snapdragon execution → validate latency → integrate automotive-style control interface